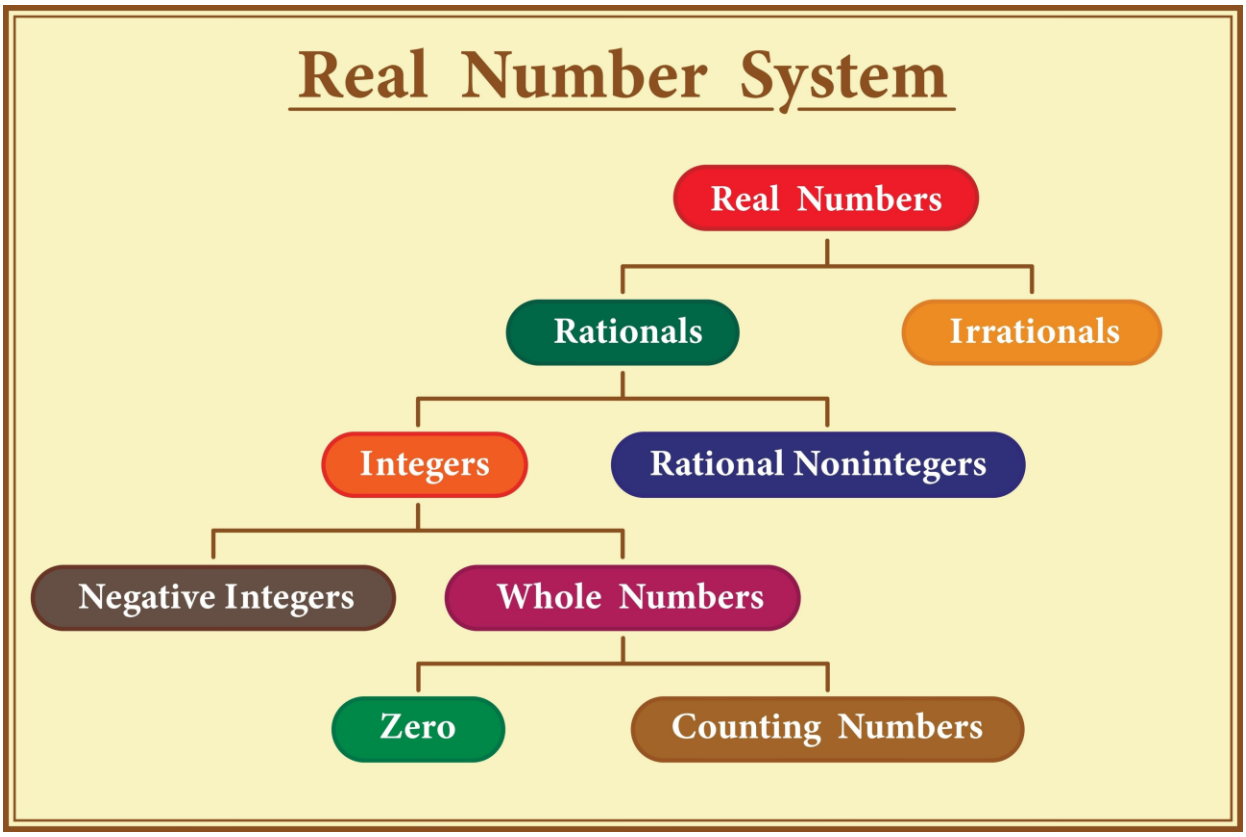
2. Real Numbers

1. The Number Family (Types of Sets)

Before we start, let's remember the groups of numbers we already know. It's like a set of boxes where each bigger box contains the smaller ones.



- **Natural Numbers (N):** These are counting numbers like {1, 2, 3, 4, ...}.
- **Whole Numbers (W):** Just Natural numbers plus zero {0, 1, 2, 3, ...}.
- **Integers (I):** These include positive numbers, zero, and negative numbers {..., -3, -2, -1, 0, 1, 2, 3, ...}.
- **Rational Numbers (Q):** Any number you can write as a fraction $\frac{p}{q}$ (where q is not zero).
- **Real Numbers (R):** This is the "Big Boss" box that contains all the numbers above.

Daily Life Example:

Think of your address. You live in a Room (Natural Number), which is in a House (Whole Number), in a Street (Integer), in a City (Rational), in a Country (Real Number).

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2. Properties of Rational Numbers

When we add or multiply rational numbers (a, b, c), they follow specific rules:

Property	Addition (a+b)	Multiplication (a×b)
Commutative	Order doesn't matter: $2+3 = 3+2$	Order doesn't matter: $2 \times 3 = 3 \times 2$
Associative	Grouping doesn't matter: $(a+b)+c = a+(b+c)$	Grouping doesn't matter: $(a \times b) \times c = a \times (b \times c)$
Identity	Adding 0 changes nothing: $a+0 = a$	Multiplying by 1 changes nothing: $a \times 1 = a$
Inverse	$a + (-a) = 0$	$a \times \frac{1}{a} = 1$ (if a isn't 0)

Exam Point:

Remember the Inverse property! For addition, the inverse is the negative of the number. For multiplication, it is the reciprocal (flipping the fraction).

3. Decimals: Terminating vs. Recurring

Rational numbers can be written as decimals in two ways:

1. **Terminating:** The division ends (e.g., $2/5 = 0.4$).
2. **Non-terminating recurring:** The numbers after the decimal keep repeating forever (e.g., $17/36 = 0.47222...$, written as $0.47\dot{2}$).

How to tell without dividing?

Look at the bottom number (denominator) of the fraction. If its only prime factors are 2 or 5, it will terminate. If it has other factors, it will recur.

Converting Recurring Decimals to Fractions:

To turn $0.777...$ into a fraction:

- Let $x = 0.777...$ (or $0.\dot{7}$)
- Since only 1 digit repeats, multiply by 10: $10x = 7.777...$
- Subtract: $10x - x = 7.777... - 0.777...$
- $9x = 7$, so $x = 7/9$.

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4. Irrational Numbers

These are the "wild" numbers. Their decimals never end and never repeat a pattern.

- **Examples:** $\sqrt{2}, \sqrt{3}$, and π.
- **The Number Line:** Every single point on a number line represents a unique Real Number.

The Story of π:

If you take any circle and divide its Circumference (the distance around) by its Diameter (the distance across), you always get roughly 3.14159... This constant value is called π (pi).

5. Surds (The Roots)

If we take the nth root of a number a (written as $\sqrt[n]{a}$) and the result is an irrational number, we call it a Surd.

The Parts of a Surd:

In $\sqrt[n]{a}$:

- $\sqrt{}$ is the **Radical sign**.
- **n** is the **Order** (if no number is there, it's 2, a square root).
- **a** is the **Radicand**.

Daily Life Example:

If you are asked for the square root of 25, it's 5 (not a surd). But if you are asked for the square root of 2, you get a messy decimal (1.414...). That is a surd!

Simplest Form:

You can simplify surds by finding square factors.

- **Example:** $\sqrt{48} = \sqrt{16 \times 3} = 4\sqrt{3}$.

Like Surds:

Two surds are "like" if they have the same order and the same radicand in their simplest form (e.g., $2\sqrt{3}$ and $5\sqrt{3}$).

6. Operations on Surds

- **Addition/Subtraction:** You can only add or subtract Like Surds, just like you can only add apples to apples.
    - **Example:** $7\sqrt{3} + 29\sqrt{3} = (7+29)\sqrt{3} = 36\sqrt{3}$.
- **Multiplication/Division:** You can multiply or divide any surds of the same order.

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7. Rationalization (Making it "Clean")

Math experts don't like having surds (roots) in the bottom (denominator) of a fraction. We use a Rationalizing Factor to fix this.

Conjugate Pairs:

For a binomial surd like $(\sqrt{a} + \sqrt{b})$, its conjugate is $(\sqrt{a} - \sqrt{b})$. When you multiply them, the roots disappear!

- $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3}) = 5 - 3 = 2$.

8. Absolute Value

The Absolute Value of a number is simply its distance from zero on the number line. Distance is never negative!

- It is written as $|x|$.
- $|3| = 3$ and $|-3| = 3$.

Summary

In this chapter, we learned that:

- Real Numbers include everything from simple counting numbers to complex decimals.
- We discovered Surds, which are roots that don't result in whole numbers, and how to perform math with them using Simplification and Rationalization.
- Finally, we saw that Absolute Value always gives us a positive distance.

Key Terms

- **Rational Number:** A number that can be a fraction $\frac{p}{q}$.
- **Irrational Number:** A decimal that never ends and never repeats.
- **Surd:** An irrational root of a rational number.
- **Radicand:** The number inside the root symbol.
- **Conjugate Pair:** Two-term surds that differ only by the plus/minus sign.
- **Absolute Value:** The positive distance of a number from zero.

Exam-Oriented Questions

- **Q: Is $\sqrt{25}$ a surd?**
    - **A:** No, because $\sqrt{25} = 5$, which is a rational number. A surd must be irrational.
- **Q: State the order of the surd $\sqrt[3]{7}$.**
    - **A:** The order is 3.

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- **Q: Multiply $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$.**
    - **A:** Using the formula $(a+b)(a-b) = a^2 - b^2$, we get $(\sqrt{5})^2 - (\sqrt{3})^2 = 5 - 3 = 2$.
- **Q: Find the value of $|8 - 13|$.**
    - **A:** $|8 - 13| = |-5| = 5$.
- **Q: Write $0.\dot{6}$ in $\frac{p}{q}$ form.**
    - **A:** Let $x = 0.66...$, then $10x = 6.66...$. $10x - x = 6$, so $9x = 6$, $x = 6/9 = 2/3$.
- **Q: What are the prime factors of the denominator for a terminating decimal?**
    - **A:** The prime factors must be only 2 or 5.

Keep practicing these, and you will be a Real Numbers expert in no time!

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- Real Numbers
  - Number Sets
    - Natural numbers (N)
    - Whole numbers (W)
    - Integers (I)
    - Rational numbers (Q)
    - Real numbers (R)
  - Rational Numbers
    - Properties
      - Commutative
      - Associative
      - Identity
      - Inverse
    - Decimal Form
      - Terminating type
      - Non-terminating recurring type
    - Order relation
  - Irrational Numbers
    - Non-terminating non-recurring decimals
    - Properties of operations
    - Square root of a negative number
    - Number Pi
  - Surds
    - Definitions
      - Radical sign
      - Order of surd
      - Radicand
      - Quadratic surd
    - Simplest form
    - Similar or like surds
    - Comparison of surds
    - Operations (Addition, Subtraction, Multiplication, Division)
  - Rationalization
    - Rationalizing factor
    - Rationalization of denominator
    - Conjugate pairs
    - Binomial quadratic surds
  - Absolute Value
    - Modulus of x
    - Distance from zero
    - Never negative